1. Download Amazon Fine Food Reviews Dataset from Kaggle

First, we need to install the Kaggle CLI and download the dataset. Please make sure you have uploaded your `kaggle.json` credential file to your Colab environment or have set up your Kaggle credentials.

```
#MOHANASUNDARAM
#241801169
import os
!kaggle datasets download -d snap/amazon-fine-food-reviews
!unzip -q amazon-fine-food-reviews.zip -d amazon_reviews
```

```
Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.2), please cons
Dataset URL: https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews
License(s): CC0-1.0
Downloading amazon-fine-food-reviews.zip to /content
100% 242M/242M [00:03<00:00, 83.7MB/s]
```

2 & 3. Install, Import Required Libraries and NLTK Resources

```
import pandas as pd
import string
import nltk

# Download necessary NLTK packages
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('averaged_perceptron_tagger')
nltk.download('maxent_ne_chunker')
nltk.download('words')

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk import pos_tag, ne_chunk
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger.zip.
[nltk_data] Downloading package maxent_ne_chunker to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping chunkers/maxent_ne_chunker.zip.
[nltk_data] Downloading package words to /root/nltk_data...
[nltk_data]   Unzipping corpora/words.zip.
```

4. Load the Dataset

We will load the dataset, select the `Text` column, drop missing values, and limit the dataset to 10,000 reviews for efficient processing.

```
# Load dataset
df = pd.read_csv('amazon_reviews/Reviews.csv')

# Select the column containing review text
reviews_df = df[['Text']].copy()

# Remove any missing or null entries
reviews_df = reviews_df.dropna()

# Limit the dataset to 10,000 reviews
reviews_df = reviews_df.head(10000)

display(reviews_df.head())
```

	Text
0	I have bought several of the Vitality canned d...
1	Product arrived labeled as Jumbo Salted Peanut...
2	This is a confection that has been around a fe...
3	If you are looking for the secret ingredient i...
4	Great taffy at a great price. There was a wid...

5 & 6. Preprocessing, Tokenization, and Stopwords/Non-alphabetic Removal

```
stop_words = set(stopwords.words('english'))

def clean_and_tokenize(text):
    # 5a. Lowercase
    text_lower = text.lower()

    # 5b. Remove punctuation
    text_no_punct = text_lower.translate(str.maketrans('', '', string.punctuation))

    # 6. Tokenization
    tokens = word_tokenize(text_no_punct)

    # 6. Remove stopwords and non-alphabetic tokens
    cleaned_tokens = [w for w in tokens if w.isalpha() and w not in stop_words]
    return cleaned_tokens

reviews_df['Cleaned_Tokens'] = reviews_df['Text'].apply(clean_and_tokenize)
display(reviews_df[['Text', 'Cleaned_Tokens']].head())
```

```
---------------------------------------------------------------------------
LookupError                               Traceback (most recent call last)
/tmp/ipykernel_2589/698010718.py in <cell line: 0>()
     15     return cleaned_tokens
     16
---> 17 reviews_df['Cleaned_Tokens'] = reviews_df['Text'].apply(clean_and_tokenize)
     18 display(reviews_df[['Text', 'Cleaned_Tokens']].head())

                                 11 frames
lib.pyx in pandas._libs.lib.map_infer()

/usr/local/lib/python3.13/dist-packages/nltk/data.py in find(resource_name, paths)
    577     sep = "*" * 70
    578     resource_not_found = f"\n{sep}\n{msg}\n{sep}\n"
--> 579     raise LookupError(resource_not_found)
    580
    581

LookupError:
**********************************************************************
  Resource punkt_tab not found.
  Please use the NLTK Downloader to obtain the resource:

  >>> import nltk
  >>> nltk.download('punkt_tab')

  For more information see: https://www.nltk.org/data.html

  Attempted to load tokenizers/punkt_tab/english/

  Searched in:
    - '/root/nltk_data'
    - '/usr/nltk_data'
    - '/usr/share/nltk_data'
    - '/usr/lib/nltk_data'
    - '/usr/share/nltk_data'
    - '/usr/local/share/nltk_data'
    - '/usr/lib/nltk_data'
    - '/usr/local/lib/nltk_data'
**********************************************************************
```

Next steps: Explain error

7 & 8. Parts of Speech (POS) Tagging and Named Entity Recognition (NER)

```
# Apply POS tagging
reviews_df['POS_Tags'] = reviews_df['Cleaned_Tokens'].apply(pos_tag)

# Identify Named Entity Recognition (NER)
# ne_chunk expects words with their POS tags
reviews_df['NER'] = reviews_df['POS_Tags'].apply(ne_chunk)

# Display sample results of the Information Extraction process
display(reviews_df[['Cleaned_Tokens', 'POS_Tags', 'NER']].head())
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/usr/local/lib/python3.13/dist-packages/pandas/core/indexes/base.py in get_loc(self, key)
   3804         try:
-> 3805             return self._engine.get_loc(casted_key)
   3806         except KeyError as err:

index.pyx in pandas._libs.index.IndexEngine.get_loc()

index.pyx in pandas._libs.index.IndexEngine.get_loc()

pandas/_libs/hashtable_class_helper.pxi in
pandas._libs.hashtable.PyObjectHashTable.get_item()

pandas/_libs/hashtable_class_helper.pxi in
pandas._libs.hashtable.PyObjectHashTable.get_item()

KeyError: 'Cleaned_Tokens'

The above exception was the direct cause of the following exception:

KeyError                                  Traceback (most recent call last)
                                          2 frames
/usr/local/lib/python3.13/dist-packages/pandas/core/indexes/base.py in get_loc(self, key)
   3810             ):
   3811                 raise InvalidIndexError(key)
-> 3812             raise KeyError(key) from err
   3813         except TypeError:
   3814             # If we have a listlike key, _check_indexing_error will raise

KeyError: 'Cleaned_Tokens'
```

Next steps: Explain error